

“SMART METER IS THE FOUNDATION FOR SMART GRID”



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Transmission and distribution losses in electricity are increasingly becoming a matter of great concern. According to World Resources Institute (WRI), India's electricity grid has the highest transmission and distribution losses in the world. There is a need for implementation of innovative technologies such as smart grids that save energy, reduce cost, increase reliability and transparency.

Embedded control and integrated connectivity will be at the heart of future smart grids. Freescale's solutions for smart metering and smart grid enablement address the challenge of efficient energy management and distribution.

Vivek Tyagi, country manager-sales, Freescale India, was recently in Delhi for the Metering India Event. On the occasion, he shared with Uma Bansal, deputy editor at EFY, his perspective on the metering industry, smart grid technology, need for smart meters in India and Freescale's contributions. Excerpts

What does a 'smart grid' mean?

A smart grid typically integrates technologies that augment power deployment and its conservation. Its primary purpose is to keep a complete check on the power flowing in a system through its advanced monitoring system, in order to help curb power loss, and is equipped to integrate alternative sources of electricity such as solar and wind. Smart grids also enable reduction in cost and improve reliability and accountability.

Smart grids provide consumers with detailed energy-use data, thus enabling demand-based pricing. These also allow system-wide efficient charging for hybrid and electric vehicles—an expected high-growth area that will require smart grid infrastructure.

Where in the world has it been deployed?

As of October 2006, eight International Energy Association (IEA) member countries had already signed up a five-year agreement for international cooperation and collaboration to develop more energy-efficient and resilient electricity transmission and distribution networks. This number could have easily gone up today with countries like China and India already moving to the next generation of networks and generation sources.

What's the case for India?

According to WRI, India's electricity grid has a whopping 27 per cent transmission and distribution losses. Vari-

ous Indian government agencies put this number at 30 per cent, 40 per cent or even greater than 40 per cent. This is mainly attributed to technical losses (grid's inefficiencies) and theft.

These are alarming figures and need to be addressed for the Ministry of Power's target of achieving 'Power for All' by 2012.

What all components does the smart grid use?

The first component is a smart meter that can supply real-time data. Second is a data concentrator that collects the real-time data from a few hundred metres and then transmits over any medium to a central server. Third is a central server that collects and presents the real-time data to utility authorities and allows them to make smart decisions.

Smart meter is an integral component and the foundation for smart grid, driving the need for higher performance and memory.

What's so 'smart' about smart meters?

Smart meters are gaining momentum as the world's leading economies overhaul outdated power grids and build out a smart grid that will improve how power is deployed and conserved. These address the challenge of efficient energy management and distribution, record consumers' electricity usage in more detail than traditional meters, allow automated and remote meter reading within a specified